Completeness of the Milky Way globular-cluster catalogue from the phases of cluster orbits

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Abstract

We estimate how the completeness of the Milky Way globular-cluster (GC) catalogue depends on absolute magnitude and heliocentric distance, using the 165 GCs with full six-dimensional phase-space data. The estimator conditions on each cluster's own orbit: if clusters are phase-mixed, the present position of a detected cluster samples its orbit with probability proportional to the completeness C at that point. Maximising the resulting likelihood needs no model of the intrinsic GC density. On mock catalogues with known selection it recovers completeness without bias and with calibrated intervals (recovered/true = 0.97–0.99, 68% coverage 0.67–0.71). A fixed-target orbit-superposition estimator in the spirit of Khoperskov et al. (2025) is biased high by $\approx 25\%$ on the same mocks. Applied to the data with C a function of extinction-free apparent magnitude V, completeness falls from 1 at $V < 6$ (assumed) to 0.14 at $V > 11$. Weighting each cluster by the inverse of its orbit-averaged completeness turns 165 observed GCs into 483 (68%: 335–765); the faint clusters ($M_V > -6.5$) rise from 55 to 267 (168–473). Mocks show these totals are lower limits, because faint clusters on orbits that never enter the detectable volume cannot be recovered by reweighting observed orbits. Adding extinction along every orbit from 3D dust maps (DECaPS, Bayestar19, Marshall, with SFD beyond their reach) improves the fit by $\Delta\mathrm{AIC} \approx 8$ and gives a corrected total of 411–455 (faint: 197–218). A simpler latitude proxy, completeness $C_V(V)\, g(|b|)$, gives 520. A continuous $C(V)$ with a falling faint tail fits better still, and with a $1/V_{\max}$ correction over the orbit population, which assumes faint and bright clusters share one orbit distribution, it recovers the true total within 0.9–1.1 in mocks. On the data it gives 393–418 GCs (68% intervals 274–625), of which 177–189 are fainter than $M_V = -6.5$.

1 Data

We use the 165 confirmed GCs with full phase-space data (positions, distances, proper motions, line-of-sight velocities and their errors) from the PI's compilation of Vasiliev & Baumgardt (2021) astrometry. Absolute magnitudes come from the Baumgardt combined table[1] as $M_V = 4.83 - 2.5\log_{10}(M_{\rm dyn}/\Upsilon_V)$, with $M_{\rm dyn}$ the dynamical mass and Υ_V the V-band mass-to-light ratio. Two confirmed clusters without 6D data are excluded. Progenitor labels (in-situ: main bulge and main disc, 61 GCs; accreted: Gaia-Enceladus, Sequoia, Helmi streams, high-energy group and Sagittarius, 64 GCs) come from the catalogue column `PROG_MA`; we have not verified their provenance (they appear to follow Massari et al. 2019).

Positions and velocities are transformed to the Galactocentric frame with the astropy v4.0 solar parameters ($R_0 = 8.122$ kpc, $z_\odot = 20.8$ pc, $\boldsymbol{v}_\odot = (12.9, 245.6, 7.78)$ km s^{-1}; Astropy Collaboration 2022). Orbits are integrated with AGAMA (Vasiliev 2019) in a static axisymmetric potential (Miyamoto–Nagai disc, $6.8\times10^{10}\,M_\odot$, $a = 3.0$, $b = 0.28$ kpc; two Dehnen spheroids, $5\times10^{9}\,M_\odot$ at 1.0 kpc and $1.71\times10^{9}\,M_\odot$ at 0.07 kpc; NFW halo, $5.4\times10^{11}\,M_\odot$, $r_s = 15.62$ kpc), forward for 5 Gyr from the present and sampled at 500 equal time steps. Energy is conserved to 2×10^{-6}. No observed cluster and no orbit sample lies beyond 150 kpc.

[1] Downloaded 2026-10-09 from https://people.smp.uq.edu.au/HolgerBaumgardt/globular/combined_table.txt.

2 Method

2.1 Per-cluster phase likelihood

Let cluster i move on a known orbit, sampled at times t, with heliocentric distance $d_i(t)$. If the population is phase-mixed in a static potential, the present phase of an undetected cluster is uniform in t. Detection with probability C tilts that distribution, so the present phase of a *detected* cluster, conditional on its orbit, has density proportional to C at that phase. The log-likelihood of the observed present positions is

$$\ln \mathcal{L} = \sum_i \Big[\ln C(M_{V,i}, d_{i,\mathrm{now}}) - \ln \langle C(M_{V,i}, d_i(t)) \rangle_t \Big]. \tag{1}$$

Two properties make Eq. (1) useful. First, no model of the intrinsic GC density enters. Second, conditioning on each detected orbit removes the dependence on which orbits happen to be in the catalogue. The overall scale of C cancels, so the data constrain C only up to one constant, and only over the range of d (or V) that the observed orbits span.

We take C piecewise constant in bins, write $\theta_k = \ln(C_k/C_0)$ with the reference bin $k = 0$ fixed, and maximise Eq. (1) with L-BFGS-B ($-8 \le \theta_k \le 4$). The likelihood is concave in θ. Intervals are 68% profile-likelihood intervals ($2\Delta \ln \mathcal{L} = 1$). The last bin is open-ended.

2.2 Two parameterisations of C

$C(M_V, d)$ per luminosity class. We split the sample into three equal-count M_V classes ($\approx$ 55 GCs each) and, within each, into equal-count distance bins of at least 15 GCs (three bins of 18–19 GCs). Bins are chosen once on the observed sample and then frozen. Each class yields $C(d)/C(\text{nearest bin})$; the classes are not linked to one another.

$C(V)$. To link the classes we assume completeness depends on apparent magnitude only, $C = C(V)$ with $V = M_V + 5\log_{10}(d/10\,\mathrm{pc})$. This is a special case of $C(M_V, d)$: a faint cluster nearby and a bright cluster farther away with the same V share the same completeness. All 165 clusters then enter one fit, with $V_i(t)$ varying along each orbit. We use eight equal-count bins in observed V (20–21 GCs each) and fix the brightest bin ($V < 6.0$) as complete, $C_0 = 1$. This single anchor makes C absolute at all M_V. V excludes extinction; Sec. 4.6 tests one extinction model.

2.3 Corrected counts

Each observed cluster receives the weight $w_i = 1/q_i$, with $q_i = \langle C(V_i(t)) \rangle_t$ its orbit-averaged completeness. Corrected distributions are weighted histograms. This is the Horvitz–Thompson (1952) inverse-inclusion-probability estimator with the inclusion probability averaged over orbital phase. It corrects for incompleteness along the orbits of observed clusters; it cannot add clusters on orbit families that are never detectable. Uncertainties on corrected counts come from a cluster bootstrap (200 replicates): each replicate resamples clusters, refits C on the fixed bins, and sums the refitted weights of the resampled clusters.

2.4 Symmetry tests

If the model holds, the present position of each cluster is a draw from its orbit's C-weighted phase distribution. For three statistics — z, y (the Galactic-longitude side) and the azimuth from the Sun's direction $|\phi'|$ — we compute each cluster's randomised probability integral transform (PIT) against that distribution, pool the 165 values, and form a Kolmogorov–Smirnov statistic. The p-value comes from 500 null simulations that redraw present phases from the fitted C and refit C in every simulation, with a +1 correction.

2.5 Comparison estimator: orbit superposition with a fixed target

Following Khoperskov et al. (2025), we also fit non-negative orbit weights so that the time-averaged occupancy of the observed orbits on an $(R, |z|)$ grid reproduces a fixed target density (spherical broken power law with inner slope 2, outer slope 4, $r_b = 5$ kpc, $r_c = 0.3$ kpc; and variants), and take $C = N_{\rm obs}/N_{\rm corr}$ in each distance bin. Unlike Eq. (1), this estimator depends on the assumed target density.

3 Validation on mock catalogues

Mock populations are drawn from an isotropic quasi-spherical distribution function in the same potential, with a broken-power-law density (inner slope 2, outer slope 4, $r_b = 5$ kpc, $r_c = 0.3$ kpc), 250 GCs per realisation and a Gaussian luminosity function $M_V \sim \mathcal{N}(-7.4, 1.2)$. Each mock cluster sits at a uniformly random phase of its orbit and is detected with probability $C_{\rm true} = \{1+\exp[(M_V-(V_{\rm lim}-{\rm DM}))/0.5]\}^{-1}$, $V_{\rm lim} = 8$, so $C_{\rm true}$ is a function of V. Scenarios B–D break the assumptions: B multiplies C by a latitude term $1 - 0.7\,e^{-|b|/5^\circ}$; C adds a non-phase-mixed clump of 15 Sagittarius-like clusters; D makes C depend on M_V and d but not through V alone. Each estimator is compared with its own correct truth: for the phase likelihood, the value it converges to for an infinite sample; for the target estimator, the true detected-to-intrinsic ratio per bin.

Table 1: Mock recovery of relative completeness $C(d)/C(\text{nearest bin})$ per luminosity class (100 realisations per scenario). Ratios are recovered/true, median and 16–84% range; coverage is the fraction of nominal 68% intervals containing the truth.

Estimator	A: assumptions hold	B: latitude-dependent C	C: Sgr-like clump
Phase likelihood	0.99 (0.68–1.41); cov. 0.67	0.98 (0.71–1.52); 0.71	0.97 (0.68–1.34); 0.71
Target, plain ratio	1.29 (0.88–1.85); 0.48	1.30; 0.54	1.39; 0.45
Target, bias-reduced	1.23 (0.84–1.77); 0.50	1.22; 0.58	1.32; 0.47

Table 2: Mock validation of $C(V)$ and the corrected counts (100 realisations per scenario).

	A: $C(V)$ true	B: latitude	C: Sgr clump	D: not $C(V)$
$C(V)$ shape, recovered/true	0.92	1.08	1.01	1.08
68% coverage	0.69	0.68	0.68	0.67
Corrected total / true	0.80 (0.60–1.07)	0.56	0.71	0.87
Faint ($M_V > -6.5$): observed / true	0.15	0.12	0.15	0.64
Faint: corrected / true	0.44	0.27	0.42	0.73

Tables 1 and 2 and Figs. 1 and 2 summarise the tests. The phase likelihood recovers completeness without bias and with calibrated intervals in every scenario, including those that break its assumptions. The target estimator is biased high by about 25% even when given the correct target. The corrected totals are low: in scenario A, where the model is correct, they recover 80% of the true population and 44% of the faint clusters. The shortfall comes from faint clusters whose orbits never bring them within detectable distance; no reweighting of the observed orbits recovers them. The bootstrap intervals on the total match the scatter across realisations (half-width $\approx 26\%$ of truth, scatter 24%), so their low coverage reflects this bias, not mis-sized intervals.

Two diagnostics have little power at this sample size. The symmetry tests reject at $p < 0.05$ in 6–13% of null mocks (the $|\phi'|$ test reaches $p \leq 0.01$ in 2%) and in 2–19% of mocks that break the assumptions. The AIC comparison of $C(V)$ against free per-class $C(d)$ favours the free model in only 31% of scenario-D mocks.

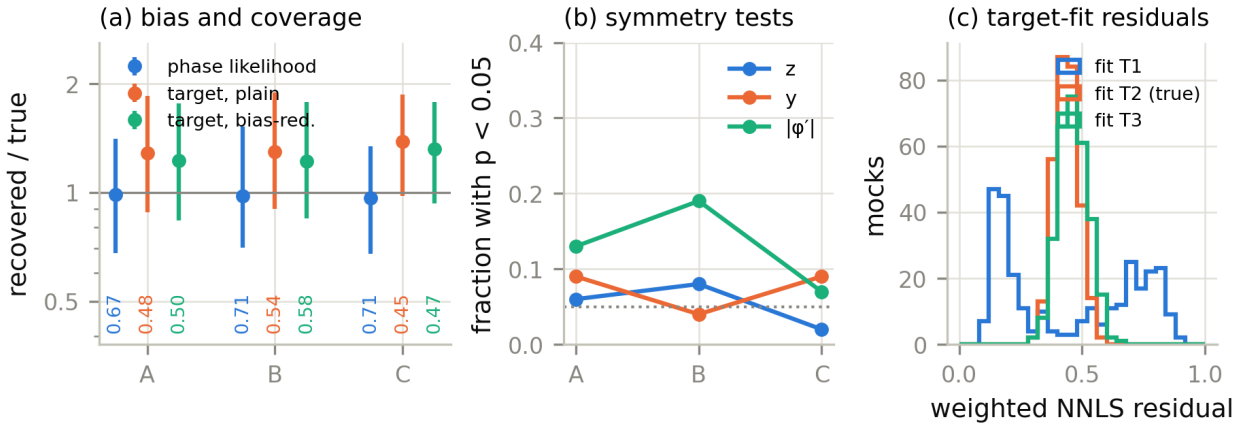


Figure 1: Mock validation of the per-class estimators. (a) Recovered/true relative completeness (median and 16–84% range) for the phase likelihood and the target estimator, plain and bias-reduced, in scenarios A–C; the vertical numbers give the 68% coverage. (b) Fraction of mocks in which each symmetry test gives $p < 0.05$; the dotted line marks 5%. (c) Target-fit residuals when the truth is T2; a residual cannot rank targets, since a T1 fit often beats the true T2.

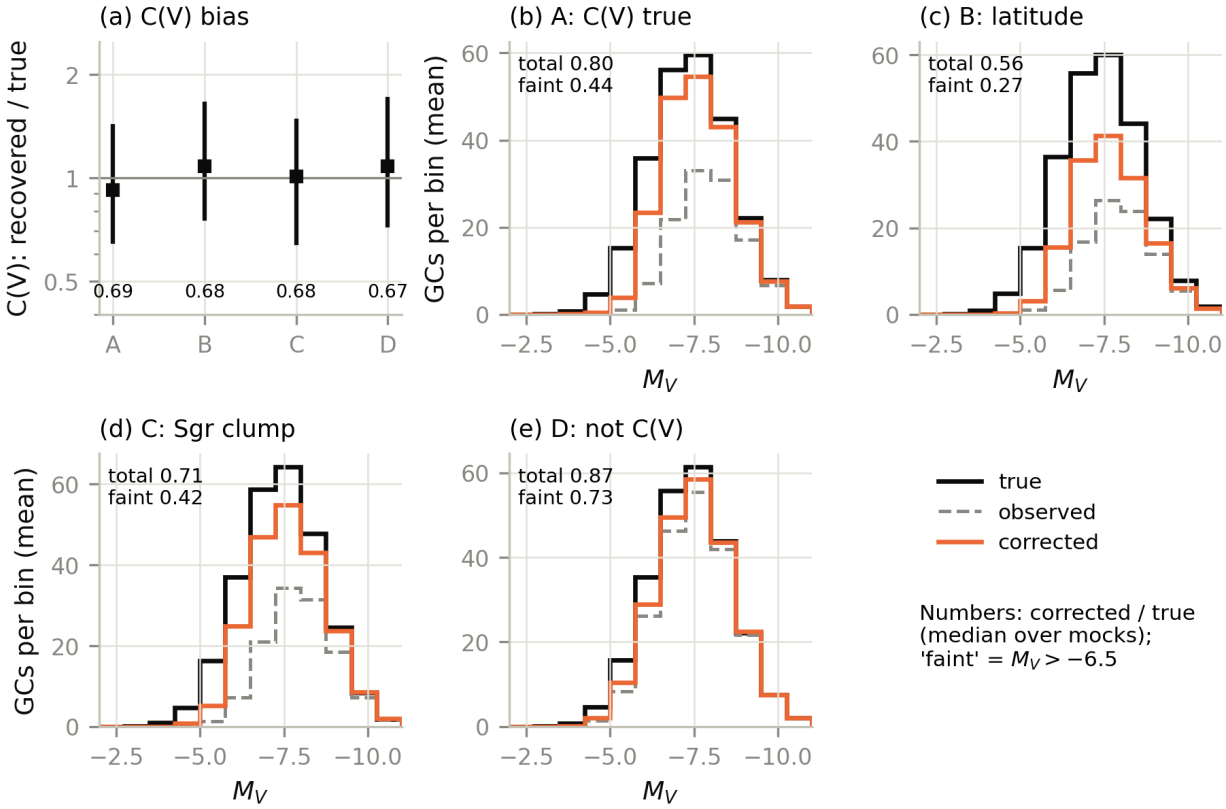


Figure 2: Mock validation of $C(V)$. (a) Recovered/true $C(V)$ per bin (median and 16–84% range) in scenarios A–D; numbers give the 68% coverage. (b)–(e) True (black), observed (dashed) and corrected (orange) M_V distributions, mean over realisations; numbers give the median corrected/true ratio for all GCs and for $M_V > -6.5$.

4 Results

4.1 Completeness per luminosity class

Table 3 and Fig. 3 give the per-class completeness. In the two brighter classes completeness stays near unity out to ≈ 10 kpc and falls to 0.4–0.6 beyond. The faint class drops to 0.44 already at 8–21 kpc. In-situ clusters lose completeness steeply with distance (0.18 beyond 8.8 kpc); accreted clusters stay near unity to 27 kpc and fall to 0.27 beyond (Fig. 4). The fixed-target estimator agrees with the phase likelihood within the intervals in most bins, but for the accreted clusters it gives values above 1 beyond 18 kpc, which the phase likelihood does not reproduce; these follow from the assumed target shape.

Table 3: Phase-likelihood completeness relative to the nearest distance bin, with 68% profile intervals. Each class holds 55 GCs in three bins of 18–19.

M_V class	nearest bin	second bin	third bin
-10.54 to -7.58	0–8.3 kpc: 1	8.3–11.1 kpc: $1.22\,[0.88, 1.71]$	>11.1 kpc: $0.63\,[0.41, 0.95]$
-7.58 to -6.50	0–7.4 kpc: 1	7.4–9.5 kpc: $0.75\,[0.52, 1.07]$	>9.5 kpc: $0.41\,[0.28, 0.61]$
-6.50 to -0.96	0–7.8 kpc: 1	7.8–21.5 kpc: $0.44\,[0.31, 0.62]$	>21.5 kpc: $0.56\,[0.27, 1.22]$

4.2 Symmetry tests

The pooled PIT tests (Fig. 5) give $p = 0.46$ for z, 0.042 for y and 0.006 for $|\phi'|$. Split by population, the in-situ clusters give $p = 0.040$, 0.29 and 0.006, and the accreted clusters 0.51, 0.49 and 0.75. Since null mocks reach $p \leq 0.01$ for $|\phi'|$ in only 2% of realisations, the in-situ clusters carry an azimuthal structure that completeness depending on (M_V, d) alone does not explain. The Galactic bar, which this analysis does not mask, is one candidate. The data do not identify the cause.

4.3 Completeness versus apparent magnitude

Table 4 and Fig. 6 give $C(V)$. Completeness falls steadily from unity at $V < 6$ to $0.14\,[0.07, 0.30]$ at $V > 11$. The per-class completeness curves implied by $C(V)$ follow the free per-class fits (Fig. 6, right), and the AIC prefers $C(V)$ (-6.0 against -1.5), although the mocks show this comparison is weak.

Table 4: $C(V)$ from the phase likelihood (brightest bin fixed at 1); 68% profile intervals. Each bin holds 20–21 GCs.

V	<6.0	6.0–6.6	6.6–7.2	7.2–7.6	7.6–8.2	8.2–9.2	9.2–11.0	>11.0
C	1	1.17	0.56	0.64	0.41	0.25	0.21	0.14
68%	–	$[0.82, 1.68]$	$[0.38, 0.82]$	$[0.43, 0.95]$	$[0.27, 0.62]$	$[0.16, 0.39]$	$[0.13, 0.35]$	$[0.07, 0.30]$

4.4 Corrected luminosity and metallicity distributions

Weighting by $1/q_i$ (weights from 0.94 to 6.9) turns the 165 observed GCs into 483 (68%: 335–765). The correction grows with faintness (Fig. 7): bright clusters ($M_V < -8$) rise from 40 to 60, intermediate ones ($-8 \leq M_V < -6.5$) from 70 to 156, and faint ones ($M_V \geq -6.5$) from 55 to 267 (168–473). The corrected metallicity distribution is about three times the observed one at all [Fe/H] and keeps its bimodal shape, with peaks near -1.5 and -0.6. Given Sec. 3, these corrected numbers are lower limits if the real selection resembles the mocks.

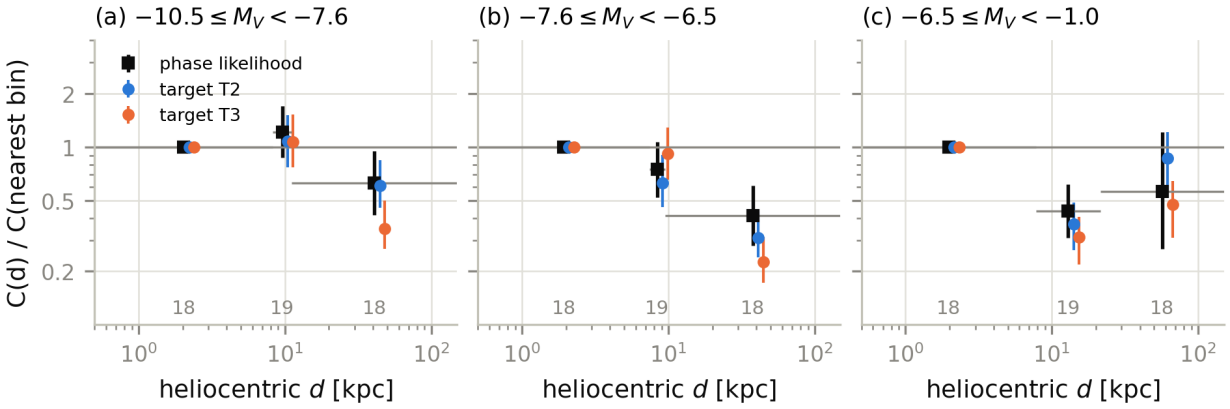


Figure 3: Completeness relative to the nearest distance bin for three M_V classes: phase likelihood (black squares, 68% profile intervals; grey bars show the bin extent) and the fixed-target estimator for targets T2 and T3 (coloured; bias-reduced, 68% bootstrap). Numbers give the observed clusters per bin.

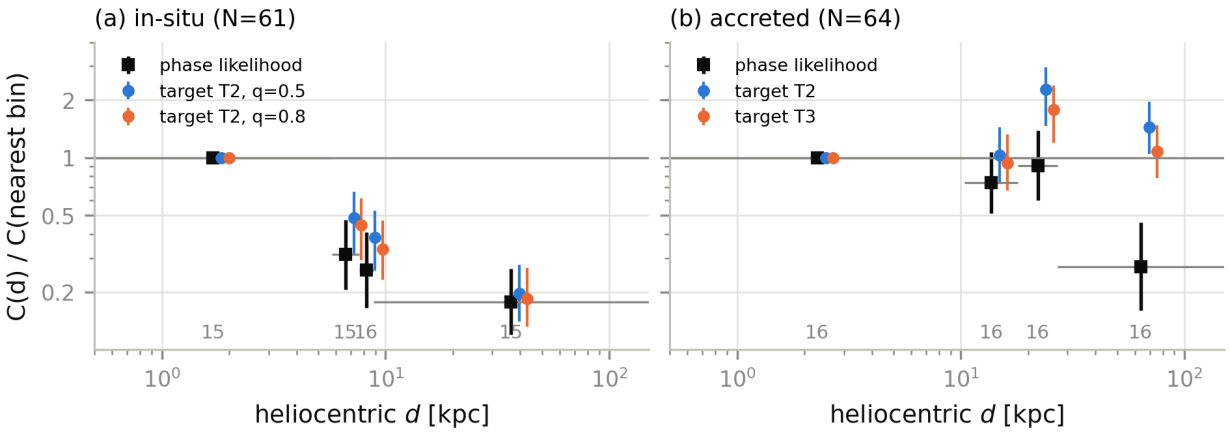


Figure 4: As Fig. 3, for in-situ (left) and accreted (right) clusters, all luminosities combined.

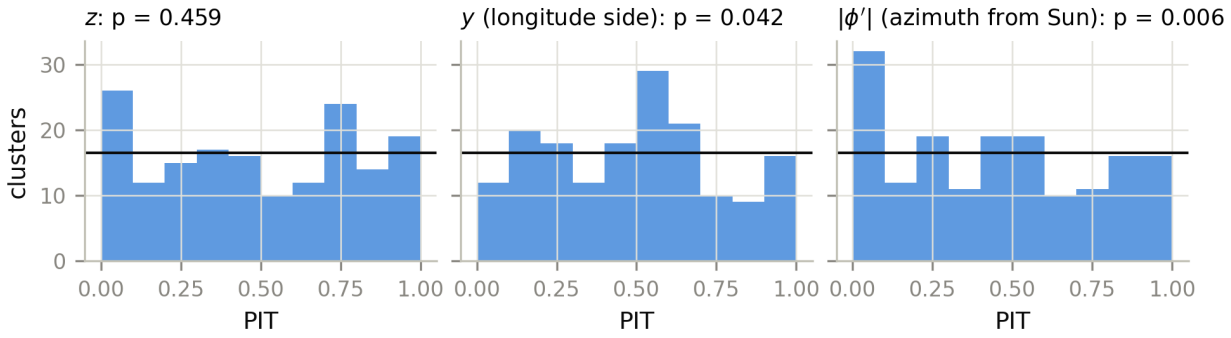


Figure 5: Pooled PIT values for z, y and $|\phi'|$ against each orbit's C-weighted phase distribution. Uniform histograms are expected if the model holds; the titles give the KS statistic and the null-calibrated p-value.

4.5 Why extinction matters, and what the mocks show

Whether a cluster is found depends on how bright it looks, and dust makes it look fainter: the relevant apparent magnitude is $V + A_V$, not V. Dust is concentrated in the Galactic disc, and most GCs live in the inner Galaxy, where many sightlines cross the disc and the bulge. The catalogue's present-day reddenings reach $E(B-V) = 6.8$, i.e. $A_V \approx 21$ mag, and half the clusters have $A_V > 1$ mag.

Extinction enters the phase likelihood in two places.

1. **The shape of C.** Along an orbit a cluster's brightness changes not only with distance but also with the sightline: passing behind the bulge it can dim by several magnitudes at almost the same distance. If A_V is left out, the fit can only attribute that dimming to distance or luminosity, which distorts $C(V)$.

2. **The corrected counts.** An orbit that spends time behind thick dust is detectable for a smaller fraction of its period. Ignoring the dust overestimates its orbit-averaged completeness q_i, so its weight $1/q_i$ is too small and the counts are under-corrected.

Because extinction depends on direction as well as distance, it also breaks the assumption that completeness depends on M_V and d alone, unless it is folded into V.

To test this we added a mock scenario (E) in which detection depends on $V + A_V$, with A_V computed along every orbit from an exponential dust disc (A_V of 1 mag kpc^{-1} in the plane at the solar circle, radial scale 3 kpc, vertical scale 0.1 kpc). We then fit each mock twice: with the true A_V added to V, and ignoring dust. Over 100 realisations:

- **The completeness curve is recovered when dust is included:** recovered/true $C(V+A_V)$ is 1.00, with 68% coverage 0.62.

- **Including dust reduces the bias in the corrected counts, but does not remove it:** the corrected total is 0.61 of the truth with the correct A_V and 0.51 ignoring dust; for faint clusters ($M_V > -6.5$) the figures are 0.18 and 0.16. The remaining shortfall is the unseen-orbit limitation of Sec. 3, which dust makes worse because it hides more orbits entirely: only 6% of the faint mock clusters are detected.

- **The likelihood can tell a correct dust model from no dust:** the AIC favours the true extinction model over ignoring dust in 98% of the realisations.

The last point matters for the real data. The likelihood comparison is a sensitive test, so the preference for no extinction over the present-day-reddening model in Sec. 4.6 is evidence against that particular model, rather than against the importance of dust. Separately, mock scenario B, whose latitude-dependent selection mimics dust that the model does not include, biases the corrected total low (0.56 of the truth).

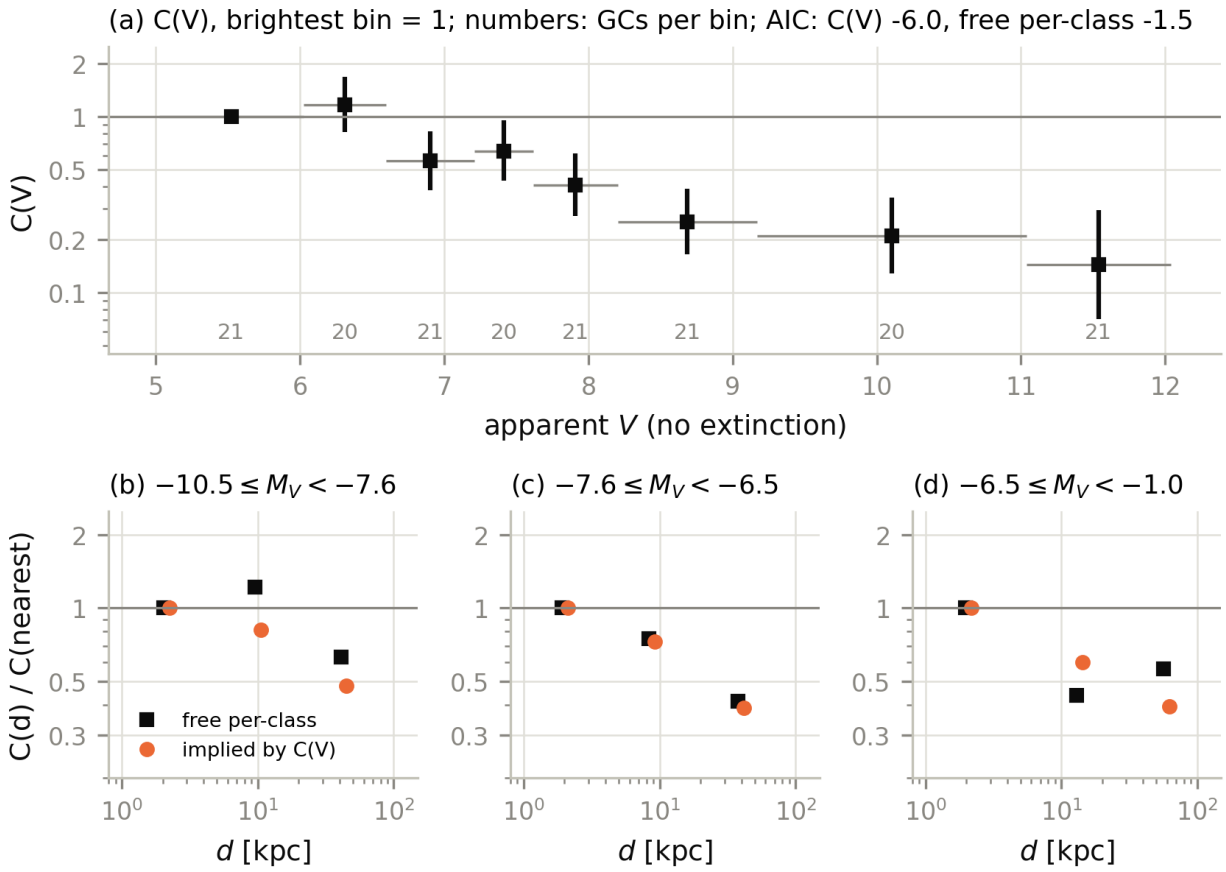


Figure 6: (a) $C(V)$ with 68% profile intervals; grey bars show the bin extent and numbers give clusters per bin. (b)–(d) Per-class completeness versus distance from free fits (black) and implied by $C(V)$ (orange).

4.6 Extinction: present-day reddening, 3D maps, and simple proxies

We tested three treatments of dust, each compared with the no-dust $C(V)$ fit by AIC (Table 5).

Present-day reddening held fixed. Adding $A_V = 3.1\,E(B-V)$, with each cluster's catalogue reddening at its present position held fixed along its whole orbit, raises the totals to 689 (457–1172) and 337 (203–651) faint. The data disfavour this model (AIC +1.3 against −6.0 without dust). Holding present reddening fixed while a cluster moves to other sightlines is unphysical; this test shows only that the total is sensitive, by about 43%, to an untested prescription.

3D dust maps along each orbit. We evaluate extinction at every one of the 165 × 500 orbit samples with the `dustmaps` package. The maps are combined by priority: DECaPS 3D (Zucker et al. 2025; southern plane $239° < l < 6°$, $|b| < 10°$, reliable to 10 kpc), then Bayestar19 (Green et al. 2019; where flagged reliable), then Marshall et al. (2006; inner plane, within each sightline's last distance slice), and SFD (Schlegel et al. 1998) otherwise. DECaPS supplies A_V for 24% of the orbit samples, Marshall for 13%, Bayestar19 for 8% and SFD for the remaining 55%, mostly at high latitude. Conversions: $A_V = 3.1\,E(B-V)$ for DECaPS and Bayestar19 ($E(B-V) = 0.883\times$ the Bayestar19 value), $A_V = 8.8\,A_{K_s}$ for Marshall, and $A_V = 2.742\,E(B-V)_{\rm SFD}$ (Schlafly & Finkbeiner 2011). Beyond a 3D map's reliable distance the true A_V is bracketed: the low bracket keeps the map's last value (no more dust), the high bracket uses the SFD total of the sightline (all its dust in front). At the clusters' present positions the maps give 5–8% less extinction than the catalogue reddening (median map/catalogue ratio 0.92 and 0.95 for the low and high brackets, 144 clusters with catalogue $A_V > 0.1$).

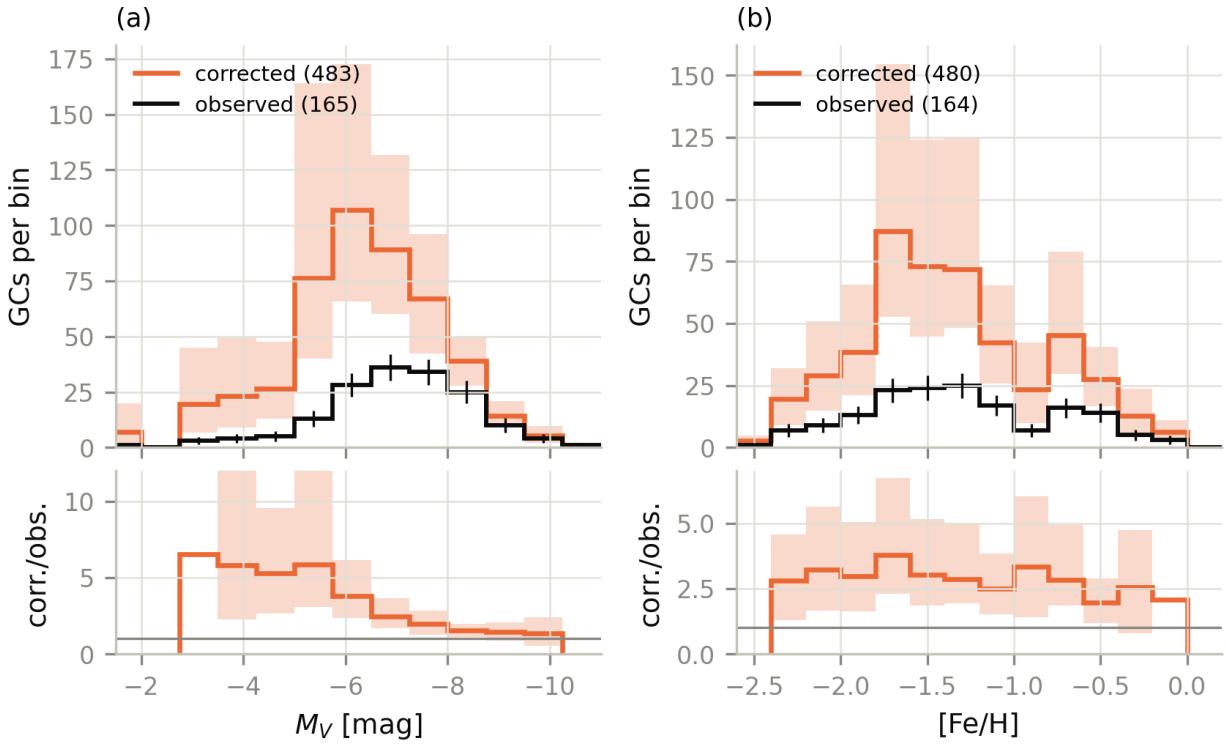


Figure 7: Observed (black) and completeness-corrected (orange, 68% bootstrap band) M_V and [Fe/H] distributions using $C(V)$. Lower panels: corrected/observed.

This model fits the data best (AIC -13.7 and -14.4 for the two brackets, against -6.0 without dust). Since the mocks show the likelihood picks the correct dust model in 98% of realisations (Sec. 4.5), the improvement is meaningful. The corrected total *falls* to 411–455, with tighter intervals, and the faint count to 197–218 (Fig. 8). Accounting for dust lets distant clusters behind it count as dimmed rather than distant, which changes the shape of C. With dust, the correction for metal-poor clusters ([Fe/H] < -1) drops from $\times 3.1$ to $\times 2.4$, while that for metal-rich clusters stays at $\times 2.6$, so the two now receive similar corrections.

Simple obscuration proxies. Without 3D maps, obscuration can be fitted directly: we write $C = C_V(V)\, g(x)$, with x either $|b|$ or the 2D SFD reddening of the sightline at each orbit sample, and g a step function in three equal-count bins anchored to 1 in the least obscured bin. The phase likelihood learns g from where along their orbits the clusters are caught. On mocks (Table 6) the proxies do no harm when there is no obscuration and reduce the shortfall when there is; the AIC prefers them over $C(V)$ in 12–15% of null mocks and in 69–71% of the dust mocks. On the real data, $g(|b|)$ fits better than no dust (AIC -10.4), with clusters in the lowest-$|b|$ third detected about half as often ($g = 0.49$), and gives a corrected total of 520; the 2D-SFD proxy is weaker (AIC -7.5; total 462).

4.7 Faint clusters on never-detectable orbits: continuous $C(V)$ and $1/V_{\max}$

Two sources of the faint-end shortfall. Mock tests isolate two causes of the low corrected counts in Table 2. With the true C, the own-orbit weights $1/q_i$ recover 87% of the total and 63% of the faint clusters (scenario A, median of 8 realisations): the rest are clusters whose q_i is nearly zero. With the fitted binned C the faint fraction falls to 41%. The last V bin is open, so the binned model holds C constant at all fainter V, where the true C falls to zero. This inflates q_i for faint clusters, whose orbits spend most of their time beyond the faintest detection. Even the best binned C (its pseudo-true value) recovers only 36% of the faint clusters.

Table 5: Real data: dust treatments compared. AIC is relative to the same data (lower is better); totals and faint counts ($M_V > -6.5$) with 68% bootstrap intervals.

Model	AIC	Corrected total	Faint
No extinction, $C(V)$	-6.0	483 (335–765)	267 (168–473)
Present $E(B-V)$ fixed along orbit	$+1.3$	689 (457–1172)	337 (203–651)
Proxy $C_V(V)\, g(E(B-V)_{\rm SFD})$	-7.5	462 (346–668)	251
Proxy $C_V(V)\, g(\lvert b\rvert)$	-10.4	520 (400–753)	269
3D maps along orbits, low bracket	-13.7	455 (354–634)	218 (158–327)
3D maps along orbits, high bracket	$\mathbf{-14.4}$	411 (319–566)	197 (140–291)

Table 6: Mocks: corrected total / true (faint in brackets), median over 100 realisations, for $C(V)$ alone and the two proxy models. In scenario E, the $C(V)$ column ignores dust; with the true A_V supplied, the total is 0.61 (faint 0.18).

Scenario	$C(V)$	$C_V(V)\, g(\lvert b\rvert)$	$C_V(V)\, g$(2D column)
A: no obscuration	0.80 (0.44)	0.83 (0.43)	0.82 (0.41)
B: latitude-dependent selection	0.56 (0.27)	0.64 (0.31)	0.65 (0.31)
E: dust disc	0.51 (0.16)	0.68 (0.19)	0.69 (0.19)

Continuous $C(V)$. We therefore also fit $\ln C$ as a piecewise-linear function of V, with nodes at the median observed V of the same eight equal-count bins. It is anchored to $C = 1$ at the brightest node and held flat at brighter V, and beyond the faintest node it continues with the last slope, constrained to be ≤ 0. The model has the same seven parameters as the binned one and is fitted with the same likelihood (Eq. 1).

$1/V_{\rm max}$ over the orbit population. We assume the orbit distribution of GCs does not depend on luminosity. Bright clusters ($M_V < -8$) are detectable along essentially all orbits, so their $1/q$-weighted orbits sample the orbit population. A cluster of absolute magnitude M placed at a random phase on that population is detected with probability

$$f(M) = \frac{\sum_{j:\, M_V<-8} q_j^{-1}\, \langle C(M+\mu_j(t))\rangle_t}{\sum_{j:\, M_V<-8} q_j^{-1}}, \qquad \mu_j(t) = 5\log_{10}\frac{d_j(t)}{10\,\mathrm{pc}} + A_{V,j}(t), \tag{2}$$

and each observed cluster gets weight $1/f(M_{V,i})$ in place of $1/q_i$. This is the $1/V_{\rm max}$ estimator (Schmidt 1968) with the survey volume replaced by the orbit population. It counts faint clusters on orbits that no observed faint cluster occupies. For the split by metallicity we apply the assumption within each class ([Fe/H] < -1, ≥ -1), using the bright clusters of the same class (31 and 9) as its orbit population.

Mock validation. Table 7 adds two scenarios whose true $C(V)$ is not log-linear in the tail: F, a shallow all-sky survey plus a deep partial-area survey, $C = 0.8\,\ell(8, 0.3) + 0.2\,\ell(13, 0.3)$ with $\ell(V_0, w) = [1 + e^{(V-V_0)/w}]^{-1}$; and G, a power-law tail, $C = (1 + V - 7)^{-2}$ for $V > 7$. The continuous model with $1/V_{\rm max}$ recovers 0.89–1.13 of the true total and 0.80–1.03 of the faint clusters in every scenario that satisfies its assumptions (A, C, D, F, G). In scenario E, with the true A_V supplied, it recovers 0.95 of the total and 0.68 of the faint clusters. It fails, as expected, under the unmodelled latitude selection of B (0.73 and 0.65). The price is variance: across realisations the total ranges over 0.67–1.55 of truth (16–84%, scenario A), and the 68% bootstrap intervals cover the truth in 53–70% of mocks, so they are somewhat too narrow. The continuous model is preferred by AIC in 55–100% of mocks.

Real data. The continuous model fits better than the binned one: AIC -22.6 against -14.4 with the map extinction (high bracket), -23.7 against -13.7 (low bracket), and -6.9 against -6.0

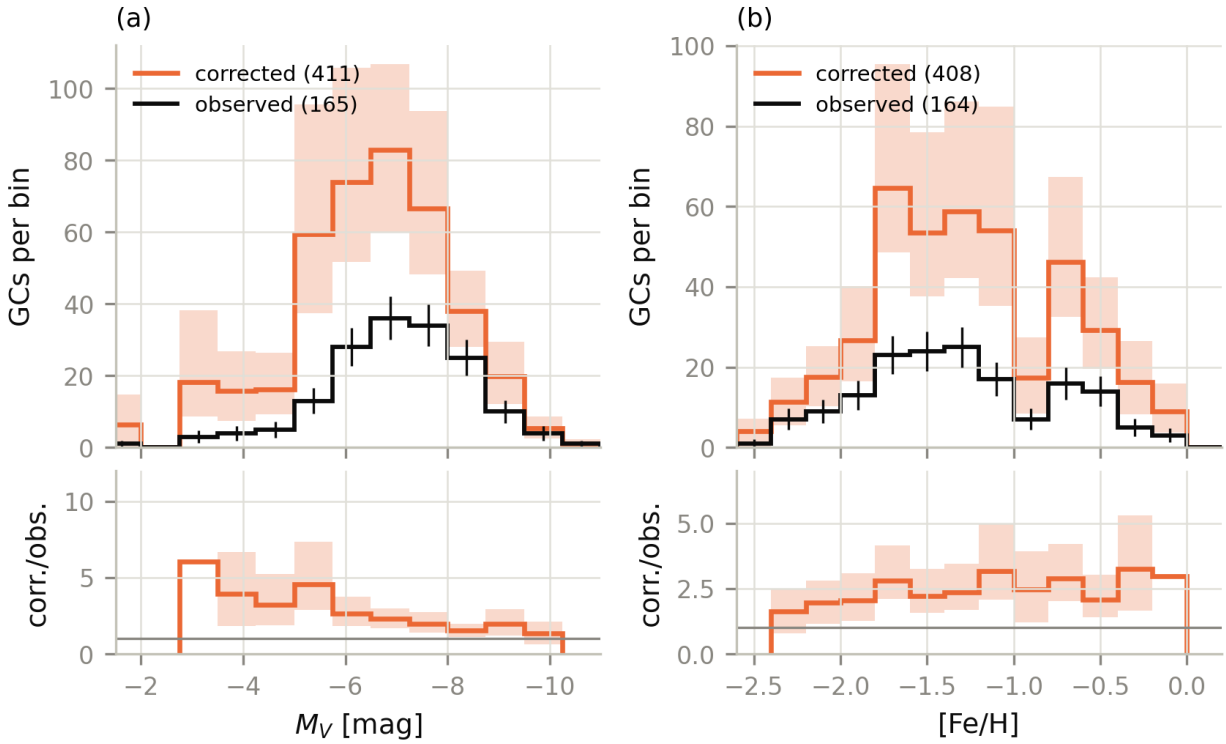


Figure 8: As Fig. 7, with A_V from 3D dust maps evaluated along every orbit (high bracket, the best-fitting binned model).

without extinction. Table 8 gives the corrected counts. With the maps, $1/V_{\max}$ gives a total of 393 (274–573) to 418 (293–625) and 177–189 faint clusters, close to the binned estimates of Table 5 (411–455; 197–218). By metallicity, the per-class $1/V_{\max}$ gives 245–259 metal-poor and 160–175 metal-rich clusters, against 289–320 and 120–133 with the binned model. The bright metal-rich clusters lie on inner, dusty orbits, so a faint metal-rich cluster on such an orbit is rarely detectable (Fig. 9).

A check of the shared-orbit assumption. If faint and bright clusters share one orbit distribution, the detected faint clusters should preferentially sit on the more detectable orbits. Then $1/V_{\max}$ should exceed the own-orbit $1/q$ for faint clusters: the ratio has median 1.0–1.4 across the mocks. On the real data it is 0.85 (maps, either bracket) and 0.61 (no extinction): observed faint clusters lie on orbits less detectable than those of bright clusters. In mocks that satisfy the assumption, a ratio below 0.85 occurs in 4–37% of realisations and below 0.61 in 3–19% (13% and 5% in scenario A), so the test does not reject the assumption at this sample size. The direction is the one expected if faint GCs are more often on extended halo orbits, in which case $1/V_{\max}$ undercounts them.

5 Limitations and next steps

1. **Unseen orbit families.** The $1/V_{\max}$ totals (Sec. 4.7) rest on orbits that do not depend on luminosity. The real data lean, not significantly, the other way. If faint GCs are preferentially on extended orbits, $1/V_{\max}$ is still a lower limit for them.

2. **Extinction.** The 3D-map treatment rests on fixed conversions (in particular $A_V = 8.8\,A_{K_s}$ for Marshall, uncertain in the inner Galaxy) and on SFD beyond the maps' reach. Map uncertainties and alternative extinction laws are not propagated; mocks with map-based rather than analytic dust would test the full chain.

Table 7: Mocks: corrected total / true (faint, $M_V > -6.5$, in brackets), median of 100 realisations. Binned: current $C(V)$ with $1/q$. Continuous: $\ln C$ piecewise linear with an extrapolated tail, with own-orbit $1/q$ or with $1/V_{\max}$. In E, the true A_V is supplied.

Scenario	Binned, $1/q$	Continuous, $1/q$	Continuous, $1/V_{\max}$
A: $C(V)$ logistic	0.80 (0.44)	1.00 (0.83)	1.01 (0.95)
B: latitude term	0.56 (0.27)	0.74 (0.52)	0.73 (0.65)
C: Sgr-like clump	0.71 (0.42)	0.96 (0.76)	1.06 (1.03)
D: C not a function of V	0.87 (0.73)	0.88 (0.72)	0.89 (0.81)
E: dust disc	0.61 (0.18)	0.85 (0.50)	0.95 (0.68)
F: two surveys	0.94 (0.79)	1.12 (1.05)	1.13 (0.98)
G: power-law tail	0.75 (0.39)	1.04 (0.92)	0.91 (0.80)

Table 8: Real data, continuous $C(V)$: corrected counts with 68% bootstrap intervals. Faint: $M_V \geq -6.5$. $1/V_{\max}$ by [Fe/H] uses the bright clusters of the same class as orbit population.

	No extinction	Maps, low	Maps, high
AIC (binned)	-6.9 (-6.0)	-23.7 (-13.7)	-22.6 (-14.4)
Total, own-orbit $1/q$	749 (467–1185)	461 (338–685)	432 (321–628)
Total, $1/V_{\max}$	563 (348–841)	418 (293–625)	393 (274–573)
Faint, own-orbit $1/q$	527 (270–910)	224 (149–345)	208 (137–314)
Faint, $1/V_{\max}$	322 (175–508)	189 (119–292)	177 (112–274)
[Fe/H] < -1, $1/V_{\max}$ per class	425 (265–645)	259 (176–346)	245 (165–327)
[Fe/H] ≥ -1, $1/V_{\max}$ per class	119 (75–169)	175 (102–277)	160 (94–251)

3. **Error propagation.** The intervals cover sampling noise only. An ensemble of perturbed phase-space coordinates, alternative potentials and longer, denser integrations would give the systematic spread.

4. **Realistic mocks and interval calibration.** The bootstrap intervals of the $1/V_{\max}$ totals cover the truth in 53–70% of mocks (Sec. 4.7). Mocks matched to the observed sample size, magnitude–distance distribution and extinction would calibrate them directly.

5. **Anchor and assumptions.** All absolute numbers assume the brightest-V bin is complete. The method assumes phase mixing in a static axisymmetric potential; the in-situ azimuthal signal shows where that may fail.

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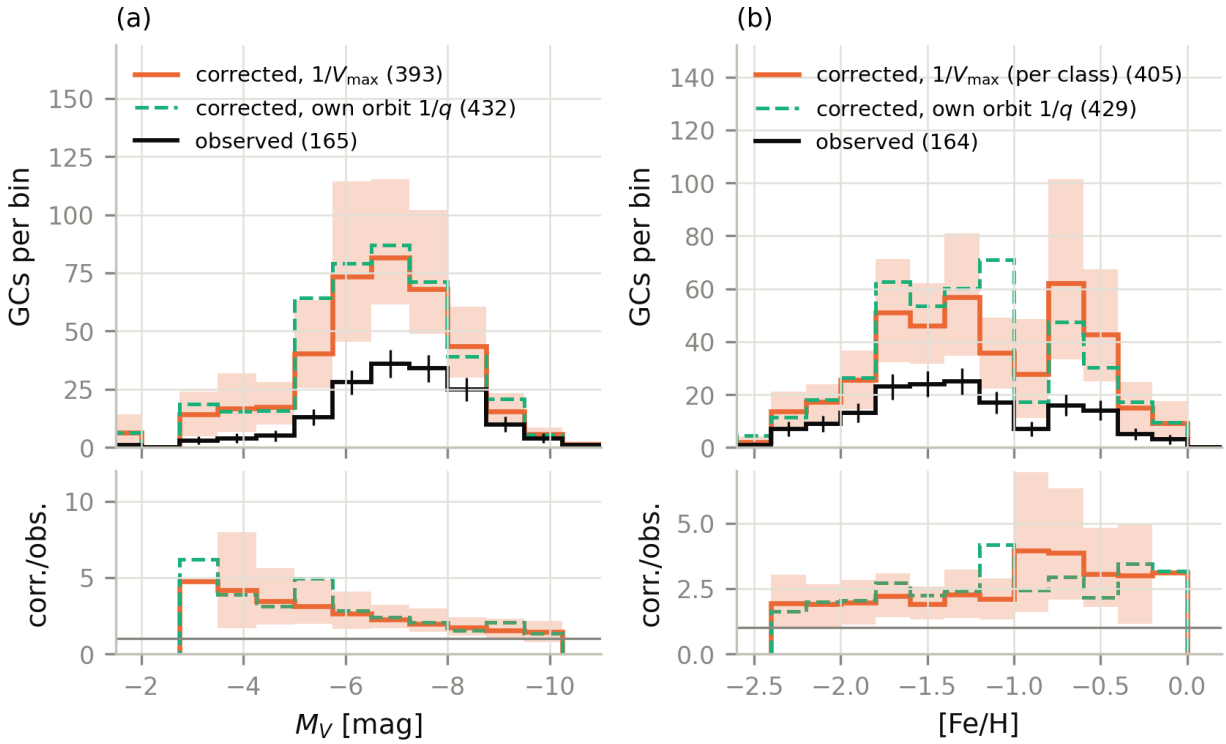


Figure 9: Corrected distributions with the continuous $C(V)$ and map extinction (high bracket): $1/V_{\max}$ (solid; band: 68% bootstrap) and own-orbit $1/q$ (dashed). (a) M_V, with the bright clusters of all metallicities as orbit population. (b) [Fe/H], with $1/V_{\max}$ computed per class.

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